

helix-core (Rust client/connector core)

helix-core (Rust client/connector core)

Revision: 1 Last modified: 2026-06-25T00:00:00Z

Master technical specification — **Volume 4 (Clients)**, nano-detail deep-dive. This document **deepens** the `helix-core` (Rust client/connector core) part of the pass-1 client overview [04_UI §0/§1/§5, 04_ARCH §5.1] into an implementation-ready specification of **the Rust workspace that the FFI consumes**: the cargo crate layout (`helix-transport`, `helix-wg`, `helix-ffi`, the orchestrator+reconciler in `helix-core`), the size/memory build strategy (`opt-level=z · LTO · panic=abort · strip`), the network-map reconciler (consume `RouteMap/MapDelta` → bring WG peers up/down, update `AllowedIPs`), the kill-switch + DNS state machine, the `---` mode=connector posture sharing the same core, and **how the core sits beneath the FFI**.

Boundary with sibling docs. This document is the *client-core* counterpart of the data-plane internals owned by Volume 2: the **orchestrator three-loop**, **the TunnelStatus broadcast**, **the reconciler diff/apply**, **the backoff machine**, and **the kill-switch/DNS coupling are specified in depth by `v02-data-plane/orchestrator-and-state.md`** (cited [v02-orch §N]) and the carrier seam by **`v02-data-plane/transport-trait.md`** (cited [v02-tx §N]). This document **owns** the *packaging + FFI seam* of that core: the cargo workspace, the per-platform library artifacts, the build/memory profile, the `helix-ffi` crate surface, and the FFI-driven *entry points* into the orchestrator/reconciler/kill-switch (it drives them; v02 defines their bodies). It does **not** re-specify the three loops, the Tunn crypto, the ladder algorithm, or the per-carrier wire formats. The FFI surface and the `TunnelStatus` enum here are **kept byte-consistent with** [v02-orch §4.1, §5] and the FFI sketch in [04_UI §5 / doc 03 §3].

Evidence base. Citations inline by id: [04_UI] = 04_VPN_CLD/HelixVPN-helix-ui-Flutter.md; [04_ARCH §N] = 04_VPN_CLD/HelixVPN-Architecture-Refined.md; [04_P0 §N] = ...Phase0-Spike.md; [research-flutter_ffi] / [research-ios_android] = the cited research digests; [v02-orch §N] / [v02-tx §N] = the Volume-2 data-plane nano-docs; [SYN §N] = cross-document synthesis. Any claim not grounded in the evidence base is tagged UNVERIFIED per constitution §11.4.6 — never fabricated.

Table of contents

- [0. Position, ownership, and invariants](#)
 - [1. The cargo workspace & crate layout](#)
 - [2. Per-platform library artifacts \(how the core ships\)](#)
 - [3. Size / memory build strategy](#)
 - [4. The `helix-ffi` crate — the surface beneath the FFI](#)
 - [5. `TunnelStatus` — core enum \$\rightleftharpoons\$ FFI projection](#)
 - [6. FFI lifecycle & the TUN-fd handoff \(the shim seam\)](#)
 - [7. The network-map reconciler beneath the FFI](#)
 - [8. Kill-switch + DNS state machine](#)
 - [9. Connector mode \(`--mode=connector`\)](#)
 - [10. Error handling & edge cases](#)
 - [11. Config knobs \(`CoreTunables`\)](#)
 - [12. Test points \(§11.4.169\)](#)
 - [13. Open decisions & cross-doc contracts](#)
 - [Sources verified](#)
-

0. Position, ownership, and invariants

`helix-core` is the **bounded-memory Rust data-plane core** shared by three consumers — Helix Access (client), Helix Connector (advertise/route), and the gateway edge — from *one* workspace [04_ARCH §5.1/§5.5, SYN §5, v02-tx §0]. On the client side it is consumed **through an FFI**: `flutter_rust_bridge` (frb) v2 emits the Dart glue, UniFFI emits the Swift/Kotlin glue where a native OS extension links the core directly [04_UI §5, research-flutter_ffi §1/§2]. This document is the nano-detail of *that core as a shippable, FFI-consumable artifact*.

0.1 What this document owns

#	Owned here	Deepened from
W1	The cargo workspace — crates, crate-type, feature flags, dependency DAG (§1)	[04_ARCH §11, SYN §6]
W2	The per-platform library artifacts (<code>staticlib/cdylib/xcframework/.so/.a/.dll</code>) and how each shim links them (§2)	[research-flutter_ffi §3, research-ios_android §2/§3]
W3	The size/memory build profile (<code>opt-level=z · LTO · codegen-units=1 · panic=abort · strip · iOS build-std</code>) + budgets (§3)	[research-ios_android §1/§2, 04_P0 §6]
W4	The helix-ffi crate surface — <code>api.rs</code> (<code>start/stop/status_stream/exits/set_exit/set_shields/advertise/attach_tun/detach_tun</code>), the mirrored value types (§4)	[04_UI §5, doc 03 §3, v02-orch §2.1]
W5	The TunnelStatus projection — core 5-variant (v02) $\rightleftharpoons$ Dart-facing FFI enum, the <code>StreamSink</code> pump, receiver contract (§5)	[v02-orch §4.1, doc 03 §3.1]
W6	The FFI-driven entry points into the orchestrator/reconciler/kill-switch — <code>attach_tun</code> , <code>apply_map</code> ,	

#	Owned here	Deepened from
	set_shields — and the C-ABI TUN-fd seam (§6, §7, §8)	[v02-orch §2.1/§6/§8, research-ios_android §3]
W7	The --mode=connector packaging — one binary, mode flag, headless daemon endpoint (§9)	[v02-orch §9, 04_ARCH §5.4]

0.2 What this document does NOT own

- The orchestrator three loops, the backoff machine, the broadcast sender body — [v02-orch §3/§4/§7]. This doc drives them through helix-ffi.
- The Transport trait, dial(), per-carrier wire formats — [v02-tx §2/§4]. The core holds opaque Box<dyn Transport> handles.
- WireGuard crypto / the Tunn state machine (helix-wg) — doc 01 §4.
- The Dart helix_core_ffi wrapper, Riverpod, TunnelPlatform Dart contract, and every pixel of UI — doc 03 §3.2/§4/§8 [04_UI §4/§5/§6].
- The WatchNetworkMap protobuf that *delivers* the RouteMap — doc 02. The core consumes the map's *shape*; a file-watch on map.json is the Phase-0 stand-in [04_P0 §10].

0.3 Invariants this document inherits and enforces at the FFI seam

#	Invariant	Source	FFI-seam enforcement
F-I1	The core owns truth about protection state; the UI is a pure function of the status stream.	[04_UI §4.2 CI2]	The FFI exposes <i>only</i> status_stream(); there is no is_connected() poll getter (§5.4).
F-I2	The core never opens the OS tunnel. The shim creates the TUN and hands the core a packet fd / pump via attach_tun (the one place crypto+transport live stays in Rust).	[04_UI §5 CI1, v02-tx I4]	start() does logic only; attach_tun(fd) is the sole packet-path entry (§6).
F-I3	One core, three consumers, no forked bodies — client/connector differ only by Mode.	[v02-orch O-I13]	CoreMode is a start() parameter; §9 adds no second loop body.
F-I4	Fail-closed by construction — kill-switch/DNS side effects are a function of OrchState transitions, surviving crash.	[v02-orch O-I11]	The FFI cannot open the firewall directly; only a state transition does (§8).
F-I5	No-logging / no-secret-leak by construction.	[v02-tx I5, §11.4.10]	SecretBytes redacts+zeroizes; the FFI never returns a secret; the status stream carries only aggregate state (§5.5).

#	Invariant	Source	FFI-seam enforcement
F-16	iOS-NE memory ceiling is the hardest budget and the reason the core is Rust not Go; every build/runtime decision defers to it.	[04_ARCH §5.6 CI4, research-ios_android §1]	§3 build profile + §3.3 runtime budget.

flowchart TB

```

    subgraph dart["Dart / UI process (doc 03)"]
        UI["helix_core_ffi wrapper + Riverpod"]
        FRB["frb_generated.dart"]
        UI --> FRB
    end
    subgraph ffi["helix-ffi crate (THIS DOC §4)"]
        API["api.rs - start/stop/status_stream/\nexits/set_exit/set_shields/advertise/attach_tun"]
        PROJ["TunnelStatus projection (§5)"]
        API --> PROJ
    end
    subgraph core["helix-core crate (v02 owns bodies)"]
        ORCH["Orchestrator (3 loops + state machine)"]
        REC["reconciler (RouteMap diff/apply)"]
        KS["kill-switch + DNS side effects"]
        ORCH --> REC & KS
    end
    subgraph deps["workspace member crates"]
        TX["helix-transport\nBox<dyn Transport>"]
        WG["helix-wg\nWgPeer / Tunn"]
        TUN["helix-tun\nTUN fd / veth+NAT"]
        ROUTE["helix-route\nCompiledPolicy"]
        DAITA["helix-daita (optional)"]
    end
    subgraph shim["per-OS native shim (doc 06)"]
        SH["NE / VpnService / wg-nt / tun\nUniFFI/JNI/C-ABI"]
    end
    FRB -- "frb v2 (Dart glue)" --> API
    SH -- "UniFFI/JNI/C-ABI: attach_tun(fd)" --> API
    API --> ORCH
    ORCH --> TX & WG & TUN & ROUTE & DAITA
    PROJ == "StreamSink<TunnelStatus>" ==> FRB

```

1. The cargo workspace & crate layout

helix-core/ is a **single cargo workspace** (snake_case, flat submodule per §11.4.28/29/74) [v02-tx §0, SYN §6/§9]. Each member is decoupled and independently testable; only helix-ffi and the binary depend on the OS/FFI.

```

helix-core/                                # the Rust workspace (own
vasic-digital repo)                        # [workspace] members + shared
├─ Cargo.toml                               # [workspace] members + shared
[profile.*] (§3)

```

```

|─ upstreams/                                # install_upstreams recipes
($11.4.36)
|─ crates/
|   |─ helix-transport/                       # L2 carriers: Transport trait
+ impls [v02-tx]
|   |   |─ crate-type = ["rlib"]
|   |   |─ helix-wg/                          # WG crypto core: WgPeer
(borington wrap) [doc 01 §4]
|   |   |─ crate-type = ["rlib"]
|   |   |─ helix-tun/                         # TUN device abstraction
(client fd / connector veth+NAT)
|   |   |─ crate-type = ["rlib"]
|   |   |─ helix-route/                      # RouteMap, CompiledPolicy,
verdict map [doc 01 §6–§7]
|   |   |─ crate-type = ["rlib"]
|   |   |─ helix-daita/                      # optional L2.5 shaping
(maybenot) [doc 01 §9]
|   |   |─ crate-type = ["rlib"]
|   |   |─ helix-core/                      # ORCHESTRATOR + RECONCILER +
state machine [v02-orch]
|   |   |─ crate-type = ["rlib"]             # (the "conductor" – owns
the loops, NOT exported directly)
|   |   |─ helix-ffi/                        # THE FFI SURFACE (THIS DOC
$4)
|   |   |─ crate-type = ["staticlib", "cdylib"] # platform-
selected at build (§2)
|   |   |─ bin/
|   |       |─ helixd.rs                     # the daemon: --mode={client|
connector}, no Flutter (§9)

```

1.1 Dependency DAG (acyclic, FFI at the apex)

flowchart TB

```

FFI["helix-ffi\n(staticlib/cdylib · frb+UniFFI)"]
BIN["bin/helixd\n(--mode=...)"]
CORE["helix-core\n(orchestrator+reconciler)"]
TX["helix-transport"]
WG["helix-wg"]
TUN["helix-tun"]
ROUTE["helix-route"]
DAITA["helix-daita\n(feature: daita)"]
FFI --> CORE
BIN --> CORE
CORE --> TX & WG & TUN & ROUTE
CORE -. "feature: daita" .-> DAITA
TX --> WG
classDef leaf fill:#eef
class WG,TUN,ROUTE,DAITA leaf

```

Rules: (1) **no crate depends on helix-ffi** — the FFI is a sink, never a dependency, so the same helix-core links into the edge binary with zero FFI weight [v02-tx I4]. (2) helix-core depends only on rlib members; the crate-type distinction (staticlib/cdylib) lives *only* in helix-ffi, so the heavy linker

artifact is produced once at the apex (§2). (3) helix-daita is behind a cargo feature = "daita" (off in Phase-1) so DAITA code is not even compiled into the lean MVP staticlib [doc 01 §9].

1.2 Workspace Cargo.toml (skeleton)

```
# helix-core/Cargo.toml
[workspace]
members = ["crates/*", "bin/*"]
resolver = "2"

[workspace.dependencies]
tokio = { version = "1", features = ["rt-multi-thread",
    "net", "time", "sync", "macros"] }
tokio-util = "0.7" # CancellationToken
bytes = "1"
async-trait = "0.1"
thiserror = "1"
zeroize = "1" # SecretBytes scrubbing (§11.4.10)
boringtun = "*" # helix-wg backend (version pinned
    at Phase-0 graduation; UNVERIFIED exact)

# frb + UniFFI live ONLY in helix-ffi (§4); not workspace-wide.

[profile.release] # the lean default (§3.1) – every
    crate inherits
opt-level = "z"
lto = "fat"
codegen-units = 1
panic = "abort"
strip = true
```

2. Per-platform library artifacts (how the core ships)

The **only** crate that changes shape per platform is helix-ffi; everything below it is rlib [research-flutter_ffi §3]. Two binding generators read one Rust surface (the canonical Mullvad/WARP pattern [04_ARCH §5.1, research-flutter_ffi §2]):

Consumer	Linkage	crate-type	Generator	
Dart (all in-process FFI platforms)	linked or dlopened by the Flutter engine	staticlib (iOS/ macOS) / cdylib (Android/ Linux/ Windows)	frb v2 (flutter_rust_bridge_codegen) → frb_generated.dart + Rust glue	.a
iOS / macOS NE (Swift)	linked into the NEPacketTunnelProvider extension	staticlib → xcframework	UniFFI (Swift) + cbindgen for the raw attach_tun C-ABI	lib.xcframework

Consumer	Linkage	crate-type	Generator	
Android (Kotlin)	System.loadLibrary in VpnService	cdylib per ABI via cargo-ndk	UniFFI (Kotlin) + JNI for attach_tun(fd)	A li (a ar x8
Windows (C#)	privileged service hosts the core	cdylib	cbindgen C header (named-pipe IPC marshals the FFI verbs)	he
Linux (Dart-FFI / in-process)	DynamicLibrary.open	cdylib	frb v2	li
HarmonyOS NEXT (ArkTS)	NAPI .so, ArkTS bridge	cdylib	cbindgen / N-API (UniFFI-Kotlin not available)	li
Aurora OS (C++/Qt)	linked into the Qt backend	cdylib/ staticlib	cbindgen C ABI	li

Generator decision (D-CORE-1, surfaced per §11.4.66). frb v2 for Dart, UniFFI for native Swift/Kotlin, cbindgen C-ABI where UniFFI is thin. flutter_rust_bridge 2.12.0 is a Flutter Favorite with first-class StreamSink support — the canonical way to push the status stream Rust→Dart [research-flutter_ffi §1/§7]. uniffi-dart is community WIP and "should not be trusted" [research-flutter_ffi §2], so UniFFI is used **only** for its first-party Kotlin/Swift output where a native extension links the core directly (iOS NE, Android VpnService) [research-flutter_ffi §2/§7]. HarmonyOS/Aurora fall back to cbindgen C-ABI because neither has UniFFI-target support [research-flutter_ffi §4/§5]. **frb-crate version MUST equal the Dart-dep version** — a drift is a build-blocking finding (§12 UT/IT) [research-flutter_ffi §1 caveat].

2.1 Build commands (per target)

```
# Android: cdylib per ABI, bundled into jniLibs [research-
ios_android §3]
cargo ndk -t arm64-v8a -t armeabi-v7a -t x86_64 -o ../apps/access/
android/app/src/main/jniLibs build --release

# iOS/macOS: staticlib → xcframework, nightly build-std for the
smallest NE footprint (§3.2)
cargo +nightly build --release -p helix-ffi --target aarch64-
apple-ios \
-Z build-std=std,panic_abort -Z build-std-
features=panic_immediate_abort,optimize_for_size
# then: lipo/xcodesbuild -create-xcframework

# Linux/Windows desktop: cdylib loaded via Dart FFI DynamicLibrary
cargo build --release -p helix-ffi --target x86_64-unknown-linux-
gnu
```

3. Size / memory build strategy

The iOS Network-Extension memory ceiling is the architecture (F-I6) [04_ARCH §5.6 CI4, research-ios_android §1/§4]. The strategy has two halves: the **build profile** (smaller code → fewer mapped pages → lower resident footprint) and the **runtime budget** (bounded buffers, no per-flow growth).

3.1 Release profile (the lean default, every crate inherits)

```
# [profile.release] from the workspace Cargo.toml (§1.2)
[research-ios_android §2]
opt-level      = "z"      # smallest code. (Evaluate "s": often
                          ~similar size, faster – §3.4 measure)
lto            = "fat"    # whole-program optimization across crates
codegen-units = 1        # better cross-module size opt (slower
                          compile, accepted)
panic         = "abort"  # drops unwinding tables + landing pads →
                          smaller + sometimes faster
strip         = true     # strip symbols (Rust 1.59+), ~3–8% extra
                          reduction
```

panic = "abort" is doubly motivated: it shrinks the binary AND removes unwinding, which the FFI boundary requires anyway — **a Rust panic! must never unwind across the C-ABI into Swift/Kotlin/Dart (UB)**. The helix-ffi entry points additionally wrap their bodies in std::panic::catch_unwind-equivalent guards that convert any caught panic into a CoreError::Internal (§10) before it reaches the boundary; with panic=abort a genuine bug aborts the process rather than corrupting the FFI — the honest, safe failure [research-ios_android §2].

3.2 iOS-only: build-std for the largest reduction

For the aarch64-apple-ios target the standard library is rebuilt with size features — this drops the gimli backtrace machinery and all panic-formatting code from std [research-ios_android §2]:

```
-Z build-std=std,panic_abort
-Z build-std-features=panic_immediate_abort,optimize_for_size
```

Representative published reduction: ~43 % over the baseline release profile [research-ios_android §2, markaicode]. This is the make-or-break lever for the G3 memory gate (§3.3). **staticlib (not dylib) for iOS** — no dynamic-loading entitlement friction, and the linker dead-strips unreferenced symbols [research-ios_android §2].

3.3 Runtime memory budget & the G3 gate

```
flowchart LR
    subgraph ios["iOS NE extension PROCESS (whole-process accounting)"]
        CORE["helix-ffi staticlib\n(WG datapath + active transport)"]
        TLS["QUIC/TLS (quinn) for MASQUE"]
    end
```



```

        BUF["bounded packet buffer pool"]
    end
    CORE & TLS & BUF --> FP["phys_footprint (Mach)\n= what jetsam
accounts"]
    FP --> GATE>{"G3: steady-state RSS\nunder ceiling, ≥30%
headroom,\n30-min soak, BOTH transports"}}
```

Fact	Value	Source
Documented NE limit (iOS 15–18)	50 MiB (whole process incl. all linked libs)	[research-ios_android §1, Apple DTS]
Observed real-world jetsam kill	~15 MB on iOS 17.3.1 (root cause PENDING_FORENSICS)	[research-ios_android §1]
Dominant kill cause	memory spikes during upload-heavy throughput, not steady-state	[research-ios_android §1, sing-box #3976]
Engineering target (this spec)	steady-state ≤ ~12–15 MB working set , never plan to 50 MiB	[research-ios_android §1/§4]
The limit	undocumented, MUST NOT be hardcoded — measure on real devices	[research-ios_android §1, Quinn DTS]

Runtime rules: (1) **bounded packet-buffer pool** sized to negotiated MTU + headroom (§6.3), never an arbitrary large read buffer [research-ios_android §1]; (2) **WG-class low-state datapath** — no TCP connection tracking, minimal persistent heap [research-ios_android §1/§4]; (3) **measure phys_footprint / task_vm_info** (the value jetsam accounts against — UNVERIFIED exact Mach field, verify on device) sampled from inside the extension, plus Instruments Allocations/VM-Tracker [research-ios_android §1].

G3 (make-or-break) gate [04_P0 §6, doc 03 §5.1]: build the same helix-ffi as aarch64-apple-ios with §3.1+§3.2; drive a **1 GB transfer on a real device** (Simulator memory is *not* representative); sample the **extension process** RSS; run **plain-UDP and MASQUE separately** (QUIC buffers cost more). **Pass = steady-state under the device-enforced ceiling with ≥ 30 % headroom across a 30-minute soak, both transports.** Documented fallbacks in order (each a real product decision — D-CLIENT-1, surfaced [04_P0 §6.4, doc 03 §14]): (a) shrink buffers / cap QUIC flow-control windows in the iOS build; (b) move MASQUE off-device for iOS (plain WG + on-path obfuscation only — partial feature loss); (c) split the core so only the lean WG datapath is in-extension and QUIC negotiation lives in the app via app-extension IPC.

3.4 Android memory posture (contrast)

Android VpnService runs as a normal foreground service — **no hard 15/50 MiB jetsam cap**; standard per-app lowmemorykiller applies, far more generous [research-ios_android §3]. The severe constraint is **iOS-only**; the same lean build is used everywhere (smaller is always better), but the budget is gated on iOS.

3.5 App-size budgets (the install-size half)

Budget	Target	Means	Source
Mobile install (per ABI)	Access < ~ 15–20 MB	--split-per-abi, lean staticlib (LTO+strip), tree-shake	[04_UI §10]
Core staticlib size	record per target; smallest is the goal	§3.1/§3.2; audit fat deps (regex/formatting/async) [research-ios_android §2 bitdrift]	[research-ios_android §2]

4. The helix-ffi crate — the surface beneath the FFI

helix-ffi/src/api.rs is the **only hand-authored FFI source**; both generators read it [04_UI §5, doc 03 §3.1]. It is a thin, panic-safe adapter over the helix-core Orchestrator (§6) — it holds the live Orchestrator behind a process-global handle and translates FFI calls into orchestrator methods. The surface is **frozen at the Phase-0 → Phase-1 graduation** [04_PO §11]; its shape matches [04_UI §5 / doc 03 §3] and its TunnelStatus matches [v02-orch §4.1] via the §5 projection.

4.1 api.rs — the exported function surface

```
// helix-ffi/src/api.rs – frb v2 + UniFFI generate Dart/Swift/
// Kotlin/C bindings.
// Transport/WG/orchestrator internals are v02/doc-01's; this is
// the EXPORTED surface only.
use flutter_rust_bridge::frb;

pub struct ClientConfig {
    pub map_path_or_session:
        String, // Phase 0: path to map.json; Phase 1+: control-
               // plane session token
    pub transport: String, // "auto" | "plain" |
                          // "masque" | "shadowsocks" | "uot" | "lwo"
    pub mode: CoreMode, // Client | Connector (F-I3)
}

#[frb(mirror(CoreMode))]
pub enum CoreMode { Client, Connector }

// ---- lifecycle (LOGIC ONLY – NOT the OS tunnel; the shim owns
// that, F-I2) ----
/// Build the Orchestrator (§6.1), arm the kill-switch closed
// (§8.2), spawn the
/// loops, emit Connecting. Idempotent: a second start while
// running is Err(AlreadyStarted) (§6.1).
pub async fn start(cfg: ClientConfig) -> Result<(), CoreError>;
/// Graceful stop: cancel loops, FIN transport, revert kill-
// switch+DNS per §8.6, emit Down{stopped}.
```

```

pub async fn stop() -> Result<(), CoreError>;

// ---- the live status stream (maps to the orchestrator
//       broadcast, §5) ----
/// frb StreamSink → Dart broadcast Stream. Many concurrent sinks
/// are allowed (broadcast
/// fan-out, §5.3); each call subscribes a fresh receiver and
/// immediately re-emits the
/// cached latest status (§5.2). Returns Err(NotStarted) if the
/// core is not running.
pub fn status_stream(sink: StreamSink<FfiTunnelStatus>) ->
    Result<(), CoreError>;

// ---- exits / routing (Phase 1+) ----
pub async fn exits() -> Result<Vec<ExitOption>, CoreError>;
pub async fn set_exit(id: String, multi_hop_chain:
    Option<Vec<String>>) -> Result<(), CoreError>;

// ---- privacy shields (pure logic – no OS tunnel touched, F-I2)
// ----
pub async fn set_shields(s: Shields) -> Result<(), CoreError>;

// ---- connector mode only (§9) ----
pub async fn advertise(cidrs: Vec<String>) ->
    Result<AdvertiseResult, CoreError>;

// ---- desired-state push (Phase 1: bridges WatchNetworkMap;
//       Phase 0: file-watch) (§7) ----
pub async fn apply_map(map_json: String) -> Result<MapApplied,
    CoreError>;

// ---- shim handoff: the core NEVER opens the TUN; the shim hands
//       it a packet fd / pump (F-I2, §6) ----
pub fn attach_tun(fd: i32) -> Result<(), CoreError>; // Android
//       ParcelFileDescriptor / Linux tun fd
pub fn detach_tun() -> Result<(), CoreError>;

```

Canonical FFI contract (R7-reconciled). Every FFI verb returns the typed `CoreError` (§10.1), never anyhow: `Result` — anyhow stays *internal* to the orchestrator and is converted at the boundary via `From` impls (see v04-client/ffi-surface.md §8.1, the canonical owner of the exported surface). The Dart-facing status enum is `FfiTunnelStatus` (§5.1); `ffi-surface.md` writes it path-qualified as `ffi::TunnelStatus` purely to contrast with the orchestrator's `core::TunnelStatus` — it is **the same Rust type**.

4.2 Mirrored value types (frb #[frb(mirror)] → identical Dart/Swift/Kotlin)

```

#[frb(mirror(Shields))]
pub struct Shields {
    pub kill_switch: bool, // → KillSwitchConfig.mode
                          Strict/Off (§8.2)
}

```

```

    pub dns_protection: bool,          // →
        DnsConfig.block_plaintext_53 (§8.3)
    pub daita: bool,                  // feature-gated; Phase 2
        [doc 01 §9]
    pub post_quantum: bool,           // PQ PSK; Phase 2 [SYN §4]
    pub split_tunnel: Vec<String>,    // per-route bypass; per-APP
        split handled in the shim layer
}

#[frb(mirror(ExitOption))]
pub struct ExitOption {
    pub id: String, pub kind: String,    // "privacy_exit" |
        "network"
    pub label: String, pub country: Option<String>, pub rtt_ms:
        Option<u32>,
    pub jurisdiction: Option<String>,    // for multi-hop
        labels
}

#[frb(mirror(AdvertiseResult))]
pub struct AdvertiseResult { pub accepted: Vec<String>, pub
    conflicts: Vec<String> }

#[frb(mirror(MapApplied))]
pub struct MapApplied {                // §7 reconcile
    outcome surfaced to the UI
    pub peers_added: u32, pub peers_removed: u32, pub
        peers_changed: u32,
    pub dns_changed: bool, pub restart_free: bool,    // always
        true (0-I12); a false here is a bug
}

```

4.3 Panic-safety & the process-global handle

```

// helix-ffi/src/state.rs – exactly one live Orchestrator per
// process.
use once_cell::sync::OnceCell;
use tokio::sync::Mutex;
static RUNTIME: OnceCell<tokio::runtime::Runtime> =
    OnceCell::new();    // multi-thread runtime
static ORCH: OnceCell<Mutex<Option<helix_core::Orchestrator>>>
    = OnceCell::new();

// Every async fn body is dispatched on RUNTIME and guarded:
// - frb v2 marshals async Rust to a Dart Future; a Rust Err
//   becomes a Dart exception.
// - with panic=abort (§3.1) a genuine bug ABORTS rather than
//   unwinds across the C-ABI (safe).
// - CoreError (§10) is the structured, surfaced failure; never
//   a silent bluff (§11.4.6).

```

UNVERIFIED: whether frb v2's generated async runtime obviates a hand-rolled RUNTIME (frb manages a worker pool) — confirm at codegen; if frb owns the runtime, RUNTIME collapses to frb's [research-flutter_ffi §1]. The single-live-orchestrator invariant (one TUN, one tunnel per app process) is enforced regardless.

5. TunnelStatus — core enum $\rightleftharpoons$ FFI projection

The **core** broadcast type is the frozen 5-variant enum owned by [v02-orch §4.1]:

```
// helix-core/src/status.rs (OWNED by v02-orch §4.1 – reproduced
// for the seam, MUST match)
#[derive(Clone, Debug, PartialEq, Eq)]
pub enum TunnelStatus {
    Connecting, // dialling the
                first/seeded rung
    Handshaking, // carrier up,
                WG handshake in flight
    Connected { transport: String, rtt_ms: u32 }, // e.g.
                ("masque-h3", 23)
    Reconnecting, // a working
                tunnel dropped; re-dialling
    Down { reason: String }, // terminal for
                this attempt (§10 reasons)
}
```

The **FFI-facing** enum (the Dart/Swift/Kotlin surface, doc 03 §3.1) is a strict *projection* of the core enum plus three FFI-only additions. The projection is the **only** place the two differ; the §12 contract test asserts the mapping byte-for-byte so they can never silently drift.

```
// helix-ffi/src/api.rs
#[frb(mirror(FfiTunnelStatus))]
pub enum FfiTunnelStatus {
    Disconnected, //
                FFI-only: clean idle (pre-start / post user-stop)

    Connecting, //
                ← core Connecting

    Handshaking, //
                ← core Handshaking
    Connected { transport: String, path: String, rtt_ms:
                u32 }, // ← core Connected; `path`="direct"|"relay" (§5.1)

    Reconnecting, //
                ← core Reconnecting
    Down { reason: String }, // ←
                core Down{reason} (unexpected drop)
    Danger { kind: String }, //
                FFI-only: "leak" | "killswitch_tripped" (§5.2)
}
```

5.1 The projection map (core $\rightarrow$ FFI)

Core TunnelStatus	FFI FfiTunnelStatus	Notes
(no live orchestrator)	Disconnected	pre-start() and after

Core TunnelStatus	FFI FfiTunnelStatus	Notes
		stop() → Down{stopped} is re-mapped to Disconnected for the UI (clean idle)
Connecting	Connecting	1:1
Handshaking	Handshaking	1:1
Connected{transport, rtt_ms}	Connected{transport, path, rtt_ms}	path derived: "relay" if the active leg is DERP-style relay, else "direct" (P2P/NAT-traversal — doc 01 §8); Phase-1 default "direct"
Reconnecting	Reconnecting	1:1
Down{reason: "stopped"}	Disconnected	user-stop is clean, not a danger
Down{reason: r} (r≠stopped)	Down{reason: r}	unexpected/terminal drop (kill-switch stays closed, §8)
(kill-switch trip / leak detected)	Danger{kind}	FFI-only overlay derived from the §8 sub-machine + leak probe; drives the red stateDanger palette (doc 03 §7)

Consistency note (§11.4.6). The core never emits Disconnected or Danger — they are FFI-layer derivations, so the v02 5-variant broadcast contract is untouched. Danger is **not** a guessed UI state: it is raised only on a concrete signal — the §8 kill-switch sub-machine reporting killswitch_tripped, or an explicit DNS/leak probe firing — never inferred from prose.

5.2 The StreamSink pump

```
// helix-ffi/src/api.rs – status_stream wiring
pub fn status_stream(sink: StreamSink<FfiTunnelStatus>) {
    let mut rx = orchestrator().subscribe(); //
    broadcast::Receiver<core::TunnelStatus> [v02-orch §2.1]
    spawn_on_runtime(async move {
```

```

// emit initial Disconnected so the UI has a value before
// the first transition (CI2)
let _ = sink.add(FfiTunnelStatus::Disconnected);
loop {
  match rx.recv().await {
    Ok(core_s) => { let _ =
sink.add(project(core_s)); } // §5.1 map
    Err(broadcast::error::RecvError::Lagged(_)) =>
continue, // skip-to-latest, NOT an error [v02-orch
§4.6]
    Err(broadcast::error::RecvError::Closed) =>
break, // orchestrator gone → stream ends
  }
}
});
}

```

frb v2 turns this `StreamSink<FfiTunnelStatus>` into a Dart `Stream` that feeds a `Riverpod StreamProvider.autoDispose` [research-flutter_ffi §6]. **Riverpod 3.0 pauses the subscription when no widget listens** — so the FFI pump is torn down when the UI backgrounds the status, and re-subscribes on resume; the core keeps running regardless (the broadcast sender outlives any receiver) [research-flutter_ffi §6].

5.3 Receiver contract (Dart side, doc 03 §8.2 honors this)

Lagged MUST be treated as "re-read latest", never an error; Closed ends the stream. The status is edge-coalesced (emissions are rare — state changes), so the 64-slot broadcast ring [v02-orch §4.2] never lags unless a receiver is wedged.

5.4 No poll getter (F-I1)

The FFI exposes `no is_connected()` / `current_status()` synchronous getter. The UI's only source of protection truth is `status_stream()` — this structurally prevents the "UI says connected while the tunnel is down" bug (CI2) [04_UI §4.2].

5.5 What the FFI MUST NOT return (F-I5)

No endpoint, SSID, gateway IP, per-packet/per-flow datum, or `SecretBytes` ever crosses the FFI. `Connected.transport` is a *kind* label; `rtt_ms` is an aggregate EWMA [v02-orch §4.5, v02-tx I5]. A code-review gate (§11.4.142) rejects any FFI return embedding an endpoint or secret.

6. FFI lifecycle & the TUN-fd handoff (the shim seam)

The seam that prevents the classic VPN bug [doc 03 §3.3, v02-orch §2.1]: **lifecycle commands flow UI → shim → core** (the OS owns the tunnel process), while **status events flow core → FFI → UI** (the core owns protection truth). `set_shields/` `set_exit/exits` are pure logic and go straight UI → FFI without touching the OS tunnel (F-I2).

6.1 `start()` → builds the Orchestrator

```
// helix-ffi/src/api.rs (body sketch) – drives helix-core; v02-orch §2.1 owns Orchestrator::start
pub async fn start(cfg: ClientConfig) -> Result<(), CoreError> {
    let mut guard = orch_cell().lock().await;
    if guard.is_some() { return Err(CoreError::AlreadyStarted); } // idempotent: no double tunnel
    let initial_map =
        load_map(&cfg.map_path_or_session).await?; // Phase 0: file; Phase 1: session→WatchNetworkMap
    let oc = helix_core::OrchestratorConfig {
        mode:
            cfg.mode.into(), // F-I3
        tun:
            TunSpec::AttachLater, // the shim calls attach_tun (§6.2) – core does NOT open it
        initial_map,
        transport_policy:
            TransportPolicy::from_hint(&cfg.transport),
        kill_switch:
            KillSwitchConfig::default(), // armed-closed-until-connected (§8.2)
        dns: DnsConfig::from(&initial_map),
        tunables:
            CoreTunables::default(), // §11
    };
    *guard =
        Some(helix_core::Orchestrator::start(oc).await?); // arms kill-switch CLOSED, spawns loops, emits Connecting
    Ok(())
}
```

`start()` does **not** open the TUN. The orchestrator is built in `TunSpec::AttachLater` and idles loop A/B until `attach_tun(fd)` injects the packet pipe (F-I2). This is the contract that keeps the *only* per-platform code in the shim [04_UI §5/§6].

6.2 The TUN-fd handoff (the C-ABI seam, per OS)

sequenceDiagram
 participant UI as Flutter UI (Dart)


```

participant FFI as helix-ffi (frb/UniFFI/C-ABI)
participant SHIM as TunnelPlatform shim (native)
participant OS as OS tunnel (NE / VpnService / wg-nt / tun)
participant CORE as Orchestrator (helix-core)

```

```

UI->>SHIM: startTunnel(TunnelConfig)                %%
MethodChannel (doc 03 §4)
SHIM->>OS: request permission + create TUN (overlay IP,
routes, DNS)
OS-->>SHIM: tun fd / packetFlow
SHIM->>FFI: helix_core_start(cfg)                    %% UniFFI/
JNI/C-ABI
SHIM->>FFI: attach_tun(fd)                            %% the
single packet-path entry (F-I2)
FFI->>CORE: TunSpec::Attach(fd) → loops A/B pump
Note over CORE: tun ↔ WG ↔ transport (v02-orch §3)
CORE-->>FFI: TunnelStatus events (broadcast)
FFI-->>UI: status_stream() → ConnectButton/StatusChip
UI->>SHIM: stopTunnel()
SHIM->>FFI: detach_tun() + helix_core_stop()
OS-->>SHIM: tunnel torn down

```

Per-OS fd source [research-ios_android §3, doc 03 §5]:

- **Android (Kotlin/JNI):** `ParcelFileDescriptor.detachFd()` transfers ownership; the int fd is passed into JNI; the core `read(2)/write(2)`s directly on it. Every outbound transport socket the core opens **MUST** be `protect()`d (ideally native `protectFromVpn` to avoid per-socket JNI upcalls) so the tunnel's own packets don't loop back into the tun [research-ios_android §3].

```

// apps/access/android/.../HelixVpnService.kt
external fun coreStart(sessionToken: String): Int //
    JNI → helix_ffi start()
external fun attachTun(fd: Int): Int //
    JNI → helix_ffi attach_tun()
// ...
val pfd: ParcelFileDescriptor = builder.establish()!! //
    OS creates the TUN
coreStart(cfg.sessionToken); attachTun(pfd.detachFd()) //
    hand raw fd to the core

```

- **iOS/macOS (Swift, NE):** the extension owns `packetFlow`; there is no integer fd — instead the shim wires the read/write callbacks. The C-ABI exposes a `tun_out` push and a `set_tun_writer` callback (the core has no fd, it pumps via the closures):

```

// apps/access/ios/HelixTunnel/PacketTunnelProvider.swift
helix_core_start(cfg.sessionToken) //
    UniFFI/cbindgen
packetFlow.readPackets { packets, _ in //
    for p in packets { helix_core_tun_out(p) } //
        core encrypts + sends via transport
}
helix_core_set_tun_writer { data in //
    core → OS: decrypted IP packets

```

```

    self.packetFlow.writePackets([data], withProtocols:
        [AF_INET])
}

```

- **Linux (in-process):** the core opens/owns the tun fd directly (a setcap cap_net_admin helper or polkit-gated daemon covers privilege); the Phase-0 G5 reference platform "the core can drive the TUN directly" [04_P0 §9, doc 03 §5.4].
- **Windows:** a privileged service hosts the core + the wireguard-nt/wintun adapter; the Flutter app marshals the FFI verbs over a named pipe; the wintun ring is the core → OS packet path [doc 03 §5.3].
- **HarmonyOS NEXT (ArkTS):** the VpnExtensionAbility tun is bridged via NAPI to the .so; honest SKIP-with-reason (§11.4.3) where the SIG port lacks the VPN API [research-flutter_ffi §5, research-ios_android n/a].
- **Aurora OS (Qt/C++):** the Qt backend owns the tun; the C-ABI receives the fd [doc 03 §5.6].

6.3 The C-ABI surface for the native packet path (cbindgen)

```

// helix_ffi.h (cbindgen-generated) – the raw seam UniFFI does not
// cover (packet hot path)
int32_t helix_core_start(const char* session_token, int32_t
mode); // 0=Client 1=Connector
int32_t helix_core_attach_tun(int32_t
fd); // Android/Linux fd path
void helix_core_tun_out(const uint8_t* pkt, uintptr_t
len); // iOS push-in path (one IP packet)
void helix_core_set_tun_writer(void (*writer)(const uint8_t*,
uintptr_t)); // core → OS callback
int32_t helix_core_detach_tun(void);
int32_t helix_core_stop(void);

```

The read buffer the core allocates for tun_out/the fd read loop is **bounded to the negotiated MTU + headroom** (§3.3), never an arbitrary large value — the single strongest lever against iOS jetsam spikes [research-ios_android §1].

7. The network-map reconciler beneath the FFI

Ownership. The reconciler's diff/apply bodies are specified in depth by [v02-orch §6]. This section specifies the **FFI-driven entry** (apply_map), the consumed RouteMap/MapDelta shapes, and the *client-core* obligation to bring WG peers up/down and update AllowedIPs restart-free — kept byte-consistent with [v02-orch §6.2/§6.3].

The core is a **declarative reconciler** [04_ARCH §4.4, 04_P0 §10]: it diffs a *desired* RouteMap (Phase 1: pushed by WatchNetworkMap, doc 02; Phase 0: a file-watch on map.json) against live state and converges — peers up/down, AllowedIPs, transport policy, DNS — **without restarting** (O-I12), < 1 s convergence [04_ARCH §4.4].

7.1 Consumed shapes (frozen, from [v02-orch §6.2])

```
// helix-route (consumed by the reconciler) – MUST match v02-orch
// §6.2 byte-for-byte
pub struct RouteMap {
  pub self_overlay:
    IpAddr, // this node's overlay IP (ULA /48,
             D4)
  pub peers: Vec<PeerRoute>, // already policy-
                             filtered (need-to-know, I6)
  pub dns: Vec<IpAddr>, // tunnel DNS servers
                      (§8.3)
  pub transport_policy: TransportPolicy, // per-leg ladder
                             order/pin/budget [v02-tx §6.2]
  pub daita: Option<DaitaMachines>, // optional shaping
                             config (data, not code; feature-gated)
}
pub struct PeerRoute {
  pub wg_pubkey: [u8; 32],
  pub endpoint_candidates: Vec<SocketAddr>, // NAT traversal
                             (Phase 2)
  pub allowed_ips: Vec<IpNet>, // overlay prefixes
                             this peer is next hop for
  pub via_connector: Option<SiteId>, // 4via6 site
                             disambiguation [doc 01 §6.1]
}
pub struct MapDelta { // produced by
                      diff(), surfaced to the FFI as MapApplied (§4.2)
  pub peers_added: Vec<PeerRoute>,
  pub peers_removed: Vec<[u8; 32]>, // pubkeys to tear
                             down (revocation path)
  pub peers_changed: Vec<PeerRoute>, // allowed_ips /
                             endpoint changed
  pub dns_changed: bool, pub policy_changed: bool, pub
    transport_policy_changed: bool,
}
```

7.2 apply_map — the FFI-driven converge

```
// helix-ffi/src/api.rs – Phase 1 wires WatchNetworkMap → this;
// Phase 0 the file-watch task calls it.
pub async fn apply_map(map_json: String) -> Result<MapApplied,
  CoreError> {
  let desired: RouteMap =
    serde_json::from_str(&map_json)?; // Phase-0 stand-
    in shape == Phase-1 stream
  let delta =
    orchestrator().apply_map(desired).await?; // v02-
    orch §6.4 in-place converge (0-I12)
  Ok(MapApplied {
    peers_added: delta.peers_added.len() as u32,
    peers_removed: delta.peers_removed.len() as u32,
    peers_changed: delta.peers_changed.len() as u32,
    dns_changed: delta.dns_changed,
```

```

restart_free:
  true, // ALWAYS
  true; a false here is a bug (0-I12)
})
}

```

7.3 Apply flow (restart-free) — what the client core does per delta class

```

flowchart TD
    push["apply_map(desired) – stream event / file-watch"] --> diff["diff(actual WgRegistry/dns/policy, desired)"]
    diff --> add["peers_added → WgRegistry.insert(Tunn)\n+ install AllowedIPs route"]
    diff --> rm["peers_removed → WgRegistry.remove\n+ DROP route (revoke < 1s) [04_ARCH §4.4]"]
    diff --> chg["peers_changed → update AllowedIPs IN PLACE\n(no WG session reset if key unchanged)"]
    diff --> dns["dns_changed → re-apply tunnel DNS (§8.3)"]
    diff --> tpol["transport_policy_changed → push to ladder\n(takes effect next re-dial) [v02-orch §7.4]"]
    add --> done["return MapApplied; OrchState UNCHANGED (no restart)"]
    rm --> selfrev["removed pubkey ==\nACTIVE gateway?"]
    selfrev -- yes --> revself["tear session, close kill-switch, \nDown{auth-failed} (§8.5)"]
    selfrev -- no --> done
    chg --> done
    dns --> done
    tpol --> done

```

- **peers_added** → insert a Tunn into the WG registry and install its allowed_ips as overlay routes; the new peer is reachable without a tunnel restart [v02-orch §6.4].
- **peers_removed** → remove the Tunn and DROP its routes; if the removed pubkey is the **active** gateway (revocation-of-self), tear the session, close the kill-switch, emit Down{auth-failed} → FFI Down{"auth-failed"} (§8.5). Effective < 1 s [04_ARCH §4.4].
- **peers_changed** touching only allowed_ips → update the routing map **in place, no WG session reset** (the key is unchanged) — the "policy edit < 1 s, no restart" MVP gate (AC5) [v02-orch §6.4, 04_P1 §11.2].

7.4 Reconcile is a side task (fail-static, I3)

The reconcile source dying (stream disconnect / file-watch error) never tears the tunnel — the loops keep forwarding on the last-applied map (fail-static) [04_ARCH §2.1, v02-orch §6.5]. The FFI surface is unaffected; apply_map simply isn't called until the source recovers (Phase-1 stream re-subscribes with its own backoff, independent of the data-plane backoff).

8. Kill-switch + DNS state machine

Ownership. The kill-switch/DNS coupling bodies are specified by [v02-orch §8]. This section specifies how the **core configures and exposes** them beneath the FFI (the Shields mapping, the Danger projection) — kept consistent with [v02-orch §8.4].

The kill-switch and DNS-leak protection are **owned by the core state machine**, never hand-edited firewall rules [04_ARCH §5.6/§7]. Every open/close and DNS apply/revert is a side effect of an OrchState transition (O-I10), so there is no window where the UI shows "connected" but the firewall is open (F-I4).

8.1 Config (from Shields + RouteMap)

```
// helix-core (configured by start() from ClientConfig+RouteMap;  
// v02-orch §8.2/§8.3 owns bodies)  
pub struct KillSwitchConfig {  
    pub mode: KillSwitchMode,           // Strict |  
        Permissive | Off  
    pub allow_lan: bool,                 // permit RFC1918  
        LAN while closed (printers etc.)  
    pub allow_gateway_endpoints: Vec<IpNet>, // gateway IPs the  
        transport MUST reach (stay open)  
}  
pub enum KillSwitchMode { Strict, Permissive, Off }  
pub struct DnsConfig {  
    pub servers: Vec<IpAddr>,           // tunnel DNS (from  
        RouteMap.dns)  
    pub block_plaintext_53: bool,        // drop off-tunnel  
        :53 when Connected (leak guard)  
    pub apply: DnsApplyMethod,           // resolvconf |  
        systemd-resolved | /etc/resolv.conf | shim  
}
```

The FFI Shields (§4.2) maps onto these: kill_switch=true ⇒ KillSwitchMode::Strict; dns_protection=true ⇒ block_plaintext_53=true. set_shields() re-applies live (no restart) [doc 03 §3.1].

8.2 The side-effect matrix (from [v02-orch §8.4], drives Danger projection)

Transition	Kill-switch	DNS	FFI projection
Idle → Connecting (Strict)	install closed (allow gw endpoints only)	system DNS	Connecting
	open (egress	apply tunnel	Connected

Transition	Kill-switch	DNS	FFI projection
Handshaking → Connected	via tunnel)	DNS + block off-tunnel :53	
Connected → Reconnecting (drop)	close (O-I11)	keep tunnel-DNS block (fail-closed)	Reconnecting
* → Down{reason≠stopped}	stay closed	keep block	Down{reason} (or Danger{killswitch_tripped} if a leak probe fired)
ShuttingDown → Down{stopped} (user)	revert to off (open)	revert system DNS	Disconnected
crash / panic	closed rules remain	block remains	(process restarts; start() re-arms)

8.3 Kill-switch + DNS sub-machine (from [v02-orch §8.7])

```

stateDiagram-v2
    [*] --> Open
    Open --> Closed : Idle→Connecting (Strict) – arm before first handshake
    Closed --> OpenTunnel : Handshaking→Connected – egress via tunnel + tunnel DNS + block :53
    OpenTunnel --> Closed : Connected→Reconnecting/Down (drop) – O-I11, keep DNS block
    Closed --> OpenTunnel : re-dial succeeded (Reconnecting→Connected)
    Closed --> Open : user Stop only (§8.4 row 5) – revert system DNS
    OpenTunnel --> Open : user Stop only
    Closed --> Closed : crash (rules persist) → core restart re-arms

```

8.4 Danger projection (the FFI-only overlay, §5.2)

The FFI raises Danger{kind} when, and only when, a concrete signal fires:

- kind="killswitch_tripped" — the sub-machine reports the kill-switch closed *while the user intends connected* (a drop the user can see); or
- kind="leak" — an explicit DNS/leak probe detects off-tunnel egress.

Danger overrides any optimistic UI intent and paints the red stateDanger palette immediately (doc 03 §7/§8.4). It is never guessed (§11.4.6) — absent a signal the FFI emits Reconnecting/Down, not Danger.

8.5 auth-failed is closed-and-terminal

A revoked device (peers_removed includes the active gateway, §7.3) reaches Down{auth-failed} with the kill-switch **closed** and DNS **blocked** — a revoked client cannot leak onto the underlay by virtue of being revoked [v02-orch §8.5]. Recovery requires explicit user re-enroll through start().

9. Connector mode (--mode=connector)

Ownership. The connector forwarding/NAT/verdict-map bodies are [v02-orch §9]. This section specifies the **packaging** — one binary, the mode flag, the headless daemon, and the FFI advertise verb sharing the same core.

9.1 One binary, one core, two postures (F-I3)

The connector is the **same** helix-core Orchestrator with Mode::Connector [v02-orch §9.1, 04_P0 §4.5]. The three loops, status stream, reconciler, backoff, and transport layer are byte-for-byte identical to the client; only the routing posture differs (no default-route capture; instead advertise served CIDRs + NAT/forward decapsulated packets into the LAN) [v02-orch §9.1/§9.2]. The connector is **outbound-only** — it dials the gateway, never listens (the founding no-port-forward constraint [SYN §1]).

```
client:    [app traffic] → TUN → WG encrypt → [DAITA] → Transport
           → gateway
connector: gateway → Transport → WG decrypt → forward into
           served CIDR (and reverse)
```

9.2 The two entrypoints share the core

```
flowchart LR
    subgraph one["one helix-core workspace"]
        CORE["Orchestrator (Mode flag)"]
    end
    FFI["helix-ffi\n(Flutter Connector flavor – optional config UI)"] --> CORE
    BIN["bin/helixd --mode=connector\n(headless daemon – primary on appliances)"] --> CORE
    CORE --> POST["Connector posture:\nadvertise CIDRs · NAT/forward · verdict map (default-deny)"]
```

- **Headless daemon** (bin/helixd --mode=connector) is the **primary** form on network appliances — no Flutter, no FFI weight, just the core [04_ARCH §5.4, doc 03 §6]. It reads its RouteMap from the control plane (or map.json) and runs the same reconciler (§7) + kill-switch knobs (typically KillSwitchMode::Off — a connector is an appliance, not a privacy client [v02-orch §9.4]).

- **Flutter Connector flavor** is the **optional** config surface; it links `helix-ffi` and drives the *same* core with `CoreMode::Connector`, calling `advertise(cidrs)` for the advertise UI [doc 03 §6, §3.1].

9.3 advertise (FFI verb, connector only)

```
// helix-ffi/src/api.rs (connector mode)
pub async fn advertise(cidrs: Vec<String>) ->
    Result<AdvertiseResult, CoreError>;
// → orchestrator pushes the served CIDRs into its RouteMap; the
//    gateway reconciles them.
// AdvertiseResult.conflicts surfaces overlapping-CIDR rejections
//    to the Connector UI (doc 03 §6).
```

9.4 Status semantics on the connector

The connector emits the **same** `TunnelStatus` stream — its tunnel *to the gateway* is what `Connected{transport, rtt}` describes; `Down` means "the connector lost its gateway tunnel" [v02-orch §9.4]. The Connector Flutter flavor subscribes to `status_stream()` exactly as `Access` does (the FFI/UI code is shared, F-I3).

10. Error handling & edge cases

10.1 FFI error taxonomy

```
// helix-ffi/src/error.rs – what crosses the FFI as a Dart
//    exception / Swift-Kotlin throw
#[derive(thiserror::Error, Debug)]
pub enum CoreError
{
    canonical = ffi-surface.md §8.1
    #[error("core not started")]
    NotStarted, // status_stream/stop/apply_map/
                // exits before start
    #[error("core already started")]
    AlreadyStarted, // start() while running (§6.1)
    #[error("invalid config: {0}")]
    Config(String), // bad transport/map path; apply_map
                    // bad JSON (§7.2)
    #[error("auth failed (revoked)")]
    Auth, // maps from Down{auth-failed} (§8)
    #[error("host fatal: {0}")]
    HostFatal(String), // TUN open / firewall apply failed
                       (§8)
    #[error("tun fd invalid")]
    BadFd, // attach_tun on a dead/foreign fd
    #[error("internal: {0}")]
    Internal(String), // panic guard (§4.3) – bug,
                      // surfaced not hidden; never a secret (§11.4.10)
}
```



```
// The orchestrator's internal anyhow / OrchestratorError /
// TransportError are converted to
// CoreError at the FFI boundary via `From` impls (ffi-surface.md
// §8.1): map-parse → Config,
// tun-attach → BadFd, auth-revoked → Auth, leak-unprotectable →
// HostFatal, else → Internal.
```

Classification: HostFatal (the core refuses to run a tunnel it cannot leak-protect) → the orchestrator emits Down{host-fatal} (§8) and start() returns Err. AlreadyStarted/NotStarted are caller-discipline errors surfaced as Dart exceptions, never silent no-ops (§11.4.6). A caught panic becomes Internal (a bug, surfaced honestly); with panic=abort (§3.1) an uncatchable panic aborts the process rather than corrupting the C-ABI.

10.2 Edge cases (each maps to a §12 test point)

#	Edge case	Required behaviour	Source
C1	start() called twice	second returns Err(AlreadyStarted); one live orchestrator, one tunnel (§6.1)	§4.3
C2	attach_tun(fd) on a closed/foreign fd	Err(BadFd); no half-open pump; kill-switch unchanged	§6.2
C3	status_stream receiver wedged > 64 events behind	Lagged → skip-to-latest; core unaffected (§5.3)	[v02-orch §4.2]
C4	iOS NE memory spike during upload soak	bounded buffer pool (§3.3); G3 soak must hold ≥30% headroom; else D-CLIENT-1 fallback	[research-ios_android §1]
C5	apply_map with malformed JSON	Err(Config); last-applied map stays live (fail-static, §7.4)	§7.2
C6	Revoke (peers_removed = active gateway)	tear session, close kill-switch, FFI Down{auth-failed}; < 1 s (§7.3/§8.5)	[v02-orch §6.4]
C7	Lagged/Closed on the FFI pump	Lagged→continue; Closed→end Dart stream (§5.2)	§5
C8	Rust panic! in an FFI body	caught → CoreError::Internal; uncatchable → panic=abort process abort (never unwind across C-ABI)	§3.1/§4.3
C9	stop() then start()	clean re-arm: ladder reset to top, kill-switch re-armed closed (L-I8)	[v02-orch §5.2]
C10	frb-crate ≠ Dart-dep version	build-blocking drift finding (§12 UT)	[research-flutter_ffi §1]
C11	Android socket not protect()ed	tunnel loops its own packets back into tun — MUST protect every outbound transport socket	[research-ios_android §3]
C12			

#	Edge case	Required behaviour	Source
	HarmonyOS/Aurora VPN API absent in the fork	honest §11.4.3 SKIP-with-reason + tracked migration item; never a faked PASS	[research-flutter_ffi §4/§5]

11. Config knobs (CoreTunables)

The FFI passes through the orchestrator tunables [v02-orch §11]; the client-core adds the FFI/runtime knobs. Every numeric default tagged "derived" is the **spec recommendation**, calibrated against Phase-0 gates with captured evidence before it is frozen (§11.4.6 — a number is a guess until measured).

```
pub struct CoreTunables {
  pub orch: helix_core::Tunables,           // [v02-orch §11]:
    tick/handshake/keepalive/backoff/status
  // --- FFI / runtime ---
  pub tun_read_buf: usize,                  // bounded to
    negotiated MTU + headroom; default 1500+64 (§3.3)
    [research-ios_android §1]
  pub status_initial_emit:
    bool,                                   // emit Disconnected on subscribe; default
    true (CI2, §5.2)
  pub ios_mem_sample_period: Duration,      // phys_footprint
    sampler cadence in NE; default 1 s (derived) [research-
    ios_android §1]
}
```

UNVERIFIED: keepalive_timeout = 25 s mirrors WG's REKEY-after-no-recv window but the exact boringtun-surfaced value is a Phase-0 measurement item [v02-orch §11]. tun_read_buf is the single strongest iOS-memory lever and is measured on device, never assumed (§3.3).

12. Test points (§11.4.169)

Per §11.4.169 every workable item declares its required test types from the closed vocabulary [06-phase0-spike-wbs §0.4]; every PASS cites rock-solid captured evidence (§11.4.5/.69/.107), never config-only. The FfiTunnelStatus event trace

- the on-device RSS report are the captured evidence for the core's claims.

Test id	§11.4.169 codes	What it proves	Rig / captured evidence
HC-1	UT	the §5.1 projection map is total + correct: every core TunnelStatus → expected FfiTunnelStatus; Down{stopped} → Disconnected	table-driven unit over §5.1 mocks a (unit on)
HC-2	UT		

Test id	§11.4.169 codes	What it proves	Rig / ca evidence
		FFI $\Rightarrow$ Dart mirror parity: the generated Dart FfiTunnelStatus/Shields/ExitOption match the Rust <code>#[frb(mirror)]</code> types byte-for-byte; frb-crate==Dart-dep version	generate fixture c drift FA build
HC-3	UT	SecretBytes Debug redacts + zeroizes on drop; no FFI return embeds an endpoint/secret (F-I5)	redaction assertion grep-em proof
HC-4	IT	<code>start()</code> builds the orchestrator, <code>attach_tun(fd)</code> pumps real WG datagrams <code>tun[?]transport</code> on the netns rig; <code>curl</code> reaches LAN host	netns + boringtu plain-ud tcpdump 200
HC-5	E2E+FA	full status arc Disconnected→Connecting→Handshaking→Connected{plain-udp,direct,rtt} driven ONLY by the FFI stream; N=3 identical traces (deterministic §11.4.50)	Flutter-I toggle (window scoped M vision v (§11.4.1
HC-6	E2E	reconcile restart-free: <code>apply_map</code> edit → peer reachable, pid unchanged, <code>MapApplied.restart_free==true</code>	file-wat delta; cu newly-reachable pid unch (G6)
HC-7	E2E+PERF	<code>revoke < 1 s</code> : <code>peers_removed=active gateway</code> → session torn, FFI <code>Down{auth-failed}</code> , kill-switch closed	timestan status tra 1 s revoke—
HC-8	SEC	kill-switch closes on drop — no plaintext egress while Reconnecting/Down; off-tunnel :53 dropped when Connected	block ga mid- Connect tcpdump underlay shows Z leaked p
HC-9	MEM	G3 make-or-break : iOS NE extension RSS under the device ceiling with $\geq 30\%$ headroom, 30-min soak, plain-UDP AND MASQUE separately; upload-spike soak holds	real-dev Instrum phys_for RSS CS
HC-10	BENCH	the §3.1/§3.2 build profile produces the recorded staticlib size; <code>build-std</code> reduction measured (not assumed)	per-targ .a/.so s CSV; be after bui
HC-11	SC (stress+chaos)	<code>start/stop/attach_tun/detach_tun</code> churn N×; reconcile-source SIGKILL mid-Connected → loops keep forwarding (fail-static); no fd leak, bounded RSS	≥ 100 ch iters; RS boundec

Test id	§11.4.169 codes	What it proves	Rig / ca evidence
HC-12	CONC+RACE	concurrent FFI calls (status_stream listener + set_shields + apply_map) on one core: no torn state, no deadlock; cargo test --features loom on the handle cell	leaked f packet concurr harness. loom rep
HC-13	IT	connector posture: bin/helixd --mode=connector + the Flutter Connector flavor drive the SAME core; gateway→connector→LAN forward + reverse, ACL default-deny	netns LA allowed reachabl unautho DROPP
HC-14	SEC	Android protect()/protectFromVpn applied to every outbound transport socket — no tunnel-loopback	netns/de capture: outbound transport packets entering
HC-15	CH+HQA	a challenges/helix_qa bank entry scores PASS only on the captured status arc + leak-free capture + G3 RSS report	HelixQA autonom session, evidence (§11.4.2

Each meta-test ships a paired §1.1 mutation (e.g. flip the §5.1 projection to map Down{stopped}→Down → HC-1 FAILs; emit Connected before opening the kill-switch → HC-8 FAILs; remove the tun_read_buf bound → HC-9 must FAIL on the upload soak) so the gates provably cannot bluff [04_P0 §8, §11.4.5].

13. Open decisions & cross-doc contracts

13.1 Decisions surfaced (options + recommendation, never silently resolved — §11.4.6/§11.4.66)

#	Decision	Option A	Option B	Recommendation
D-CORE-1	FFI binding generator	frb v2 (Dart) + UniFFI (Swift/Kotlin) + cbindgen (HarmonyOS/Aurora)	cbindgen C-ABI everywhere	A — frb's StreamSink is the mature status-stream path; UniFFI gives typed Swift/Kotlin; cbindgen where UniFFI is thin [research-flutter_ffi §2/§7]
D-CLIENT-1	iOS NE memory fallback	shrink buffers / cap QUIC windows in iOS build	move MASQUE off-device or split-core app-extension IPC	Gate on the G3 measurement (§3.3); prefer A if

#	Decision	Option A	Option B	Recommendation
	(if G3 fails)			it reaches $\geq 30\%$ headroom [04_P0 §6.4]
D-CORE-2	iOS build-std adoption	nightly build-std + optimize_for_size (~43% smaller)	stable toolchain, [profile.release] only	A for the iOS target if the size win is needed to pass G3; B elsewhere (stable) [research-ios_android §2]

13.2 Cross-document contracts this document fixes

Contract	Fixed value	Consumed by
helix-ffi exported surface (start/stop/status_stream/exits/set_exit/set_shields/advertise/apply_map/attach_tun/detach_tun)	§4.1	doc 03 (Dart wrapper), every native shim
FfiTunnelStatus + its projection from the core 5-variant TunnelStatus	§5	doc 03 §3.2 (Dart mirror), [v02-orch §4.1] (must emit the 5 core variants)
TUN-fd C-ABI seam (attach_tun(fd) / tun_out / set_tun_writer / detach_tun)	§6.3	every per-platform shim (NE/VpnService/wg-nt/tun)
apply_map reconcile entry (RouteMap/MapDelta shape, restart-free)	§7	doc 02 (WatchNetworkMap source), [v02-orch §6]
Build/memory profile (opt-level=z·LTO·panic=abort·strip + iOS build-std; $\leq 12\text{--}15$ MB steady-state, G3)	§3	the iOS NE build, [04_P0 §6]
--mode=connector packaging (one binary + headless daemon, advertise)	§9	doc 03 §6 (Connector flavor), [v02-orch §9]

Sources verified

- 04_VPN_CLD/HelixVPN-helix-ui-Flutter.md [04_UI] — §0 (is/is-not), §1 (melos layout), §2 (flavors), §4 (Riverpod), §5 (FFI surface), §6 (per-platform shim matrix + HarmonyOS/Aurora), §10 (budgets).
- 04_VPN_CLD/HelixVPN-Architecture-Refined.md [04_ARCH §5] — §5.1 (two-cores strategy + frb/UniFFI), §5.3 (shim matrix), §5.4 (connector headless), §5.5 (reuse map), §5.6 (size/memory, iOS-NE = Rust-not-Go), §5.7 (Web/Aurora caveats), §11 (repo layout).

- `04_VPN_CLD/HelixVPN-Phase0-Spike.md` [`04_P0`] — §6 (iOS NE memory experiment G3 + fallbacks), §9 (FFI boundary G5 + Rust/Dart skeletons), §10 (static-map shape), §11 (graduation).
- `v09-research/research-flutter_ffi.md` [`research-flutter_ffi`] — §1 (frb v2 2.12.0 + StreamSink), §2 (UniFFI not Dart-production-ready), §3 (staticlib/cdylib packaging), §4 (Aurora fork), §5 (HarmonyOS SIG fork), §6 (Riverpod 3.0 StreamProvider pause/autoDispose), §7 (synthesis).
- `v09-research/research-ios_android.md` [`research-ios_android`] — §1 (NE memory: 50 MiB documented iOS 15–18, ~15 MB observed kills, target ~12–15 MB, do-not-hardcode), §2 (Rust size optimization: `opt-level=z/LTO/panic=abort/strip + build-std`), §3 (Android VpnService detachFd/protect/JNI), §4 (cross-platform synthesis).
- Sibling Volume-2 docs `v02-data-plane/orchestrator-and-state.md` [`v02-orch`] (Orchestrator, the 5-variant TunnelStatus, the reconciler diff/apply, the kill-switch/DNS coupling, `Mode::Connector`) and `v02-data-plane/transport-trait.md` [`v02-tx`] (the Transport trait, `SecretBytes`, invariants I1–I5). [SYN] cross-document synthesis — §5 (client architecture), §6 (repo layout), §8/§9 (submodules, constitution bindings).

Constitution: §11.4.44 (revision header), §11.4.6/§11.4.66 (decisions = options + recommendation, UNVERIFIED never fabricated), §11.4.5/§11.4.69/§11.4.107/§11.4.159 (captured/window-scoped evidence), §11.4.10 (SecretBytes no-leak), §11.4.28/.29/.74 (decoupled reusable workspace, snake_case, flat submodule), §11.4.169 (mandatory per-item test types). SPEC-ONLY — describes what to build, does not build it.

End of nano-detail specification — Volume 4 (Clients), `helix-core-rust.md`. Pairs with `v02-data-plane/orchestrator-and-state.md` (the orchestrator this FFI drives), `v02-data-plane/transport-trait.md` (the carrier seam), and doc 03 (the Dart `helix_core_ffi` wrapper + `TunnelPlatform` shim contract above this surface). The `helix-ffi` surface (§4.1), the `FfiTunnelStatus` projection (§5), the `TUN-fd` C-ABI seam (§6.3), and the `--mode=connector` packaging (§9) are the frozen Phase-0 surviving contracts [`04_P0` §11]; their implementations may evolve, their shapes may not.